

ENDPOINT COSTS AND BALANCED NEAR-MINIMIZERS UNDER GAUSSIAN REFLECTIONS

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ABSTRACT. A comparison of Euclidean costs along a finite graph of continued-fraction states depends on where the walk begins and ends as much as on the cycles it traverses. We prove a quantitative stability theorem for finite graphs carrying nonnegative integer initial, edge and terminal weights: if every cyclic component of the zero-weight subgraph has a single letter frequency, then the heavy-letter count of any balanced accepted word lies within an explicit multiple of its completed cost from that finite set of frequencies, the multiple being read off the condensation graph. Balance is what allows the finite set to replace its convex hull. The compatible slopes of a finite word are identified with the projection of a face of the arrangement by which Berstel and Pocchiola enumerated the factors of Sturmian words, which also measures how thin the arithmetic input is: of the balanced words of length n , only $O(n)$ of the roughly n^3/π^2 arise from a Markoff occurrence.

We apply the theorem to the Gaussian reflection attached to $89 = 8^2 + 5^2$, acting on primitive sources whose norm has 89-adic valuation exactly one. On an unrestricted paired-digit language this reflection never decreases subtractive-Euclidean cost, and we determine every word whose cost it preserves. Consequently every distinct reflected target of a genuine reduced Markoff source has strictly larger cost, by an amount that grows linearly with the denominator of the Farey slope outside any neighborhood of four critical slopes. An explicit infinite family of genuine sources realizes this growth. The proof of the inequality at 89 is in part computer-assisted; the stability theorem and the infinite family are not. These comparisons concern the complete norm-89 flip and do not establish global Markoff uniqueness.

1. INTRODUCTION

The Markoff equation¹

$$a^2 + b^2 + c^2 = 3abc$$

has a tree of positive integer solutions, generated from $(1, 1, 1)$ by permutations and by the replacement $(a, b, c) \mapsto (a, b, 3ab - c)$. The Markoff–Frobenius conjecture asserts that the largest entry of a solution determines the other two. It remains open, and a recurring strategy is to attach to each occurrence of a largest entry in the tree an invariant finer than the entry itself, rigid enough to separate two occurrences that share their numerical value. The invariant studied here is a cost.

Each occurrence carries a reduced fraction P/Q , its Farey slope, and determines a primitive representation of its label as a sum of two squares. The subtractive Euclidean algorithm assigns an integer cost to such a pair: the number of subtractions needed to bring the two coordinates to their common divisor, or equivalently the sum of the partial quotients of their ratio, minus one. For the occurrence pair that cost is exactly $P + Q - 2$. The natural question is whether the same integer admits a cheaper primitive

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¹We use the spelling *Markoff* for the equation and its numbers, and refer to the uniqueness question as the Markoff–Frobenius conjecture.

representation, and the comparison is subtle: a Gaussian reflection preserves the norm while changing many partial quotients at once.

Reading the continued fraction of a source and applying the reflection is a walk through a finite set of integer matrices, along which the cost difference accumulates. Nonnegative weights on the cycles of such a graph control repeated motion, but a finite walk also begins and ends somewhere, and its initial and terminal charges survive any rebalancing of the edge weights. Our first result concerns exactly that situation; the second applies it to one Gaussian prime block.

1.1. The two results. The general theorem concerns a finite graph reading one binary letter on each edge. Suppose an exact potential makes all initial, edge and terminal weights nonnegative integers, and suppose each cyclic component of the zero-weight subgraph carries a single letter frequency. Denote the finite set of those frequencies by Θ , assumed nonempty. For a balanced accepted word of length n with m heavy letters and completed cost Δ , we prove

$$\min_{\theta \in \Theta} |m - \theta n| < K(\Delta + 1), \quad (1)$$

where K is obtained explicitly from the condensation graph and from elementary count potentials inside its cyclic components. No irreducibility, determinism, or bound on the input length is required.

The role of balance is precise. An unrestricted word can spend a long time in one recurrent component and then a long time in another, so that its overall frequency lies strictly between theirs. A balanced word cannot make such a macroscopic mixture unless the two frequencies are already close. This is what turns a statement about the convex hull of Θ into the finite-set estimate (1).

Balance enters through a single compatible slope, valid simultaneously for every factor. That slope has a geometric meaning: the pairs consisting of a slope and an intercept that produce a given finite word form a convex region of the plane, and the compatible slopes are exactly its shadow on the slope axis. These regions are the faces of the arrangement through which Berstel and Pocchiola counted the finite factors of Sturmian words, so the estimate rests on one coordinate of a two-parameter picture.

Our arithmetic application fixes the Gaussian block $89 = 8^2 + 5^2$. If a primitive source has norm divisible by 89 exactly once, there is a single complementary allocation of this pair of conjugate Gaussian primes, modulo units and conjugation; we call the resulting comparison the complete norm-89 flip. For the entire language $q = \varepsilon \chi(w)2$, where χ doubles each digit, we prove that the flip cannot lower the cost, and we determine every word on which the cost is preserved:

$$q = 111\chi(s)2, \quad s \in (12111 \mid 21111)^*. \quad (2)$$

The target exchanges the two five-letter tokens. The reduced-count condition satisfied by a genuine Markoff occurrence excludes every nonempty token sequence, so every distinct genuine target has strictly larger cost.

The same finite graph has critical frequencies

$$\Theta_{89} = \{0, 1/5, 1/3, 1\}. \quad (3)$$

For a genuine source of Farey slope P/Q , the quantitative consequence is

$$\min_{\theta \in \Theta_{89}} |P - \theta Q| \leq 118\Delta + 121. \quad (4)$$

The excess therefore grows linearly with Q on every closed interval of slopes disjoint from Θ_{89} . We construct an infinite sequence of genuine sources to which this conclusion applies: their norms have 89-adic valuation exactly one, and both Farey coordinates tend to infinity.

1.2. Background and contribution. Finite transducers for homographic transformations of continued fractions originate in Raney's work, and their matrix-state interpretation continues to be useful in arithmetic applications [11, 5]. Potential transformations of weighted automata, including the accompanying changes at initial and terminal states, are standard [9]. Finiteness of an automaton carries arithmetic information in other settings as well: over a finite field, the power series that are algebraic are precisely those whose coefficients a finite automaton can generate [4]. We make no use of that equivalence here, and return to it in [section 8](#). Critical classes and the eventual concentration of optimal paths are central in min-plus spectral theory [10], and the relation between cycle frequencies and convex rotation sets has a well-developed graph-theoretic interpretation; see [3, Section 3]. We use these mechanisms explicitly and attribute them where they occur. What (1) adds is a balance condition imposed on individual finite inputs, which replaces the convex hull of the critical frequencies by the frequencies themselves.

The general statement proved here is the finite balanced-path estimate (1), with its endpoint hypotheses and an explicit condensation constant. We also record how narrow the arithmetic input is: of the roughly n^3/π^2 balanced words of length n , at most $3n + 3$ are cores of genuine occurrences. The norm-89 application adds the endpoint inequality, the complete equality language and the quantitative consequence for genuine occurrences. We make no priority claim for the underlying potential method, for the classical balance lemma, or for the correspondence between central words and Markoff numbers.

The proof of the endpoint inequality at 89 is in part computer-assisted: it rests on finitely many integer inequalities attached to a finite graph, verified in exact arithmetic. [Section 5](#) states precisely which finite assertion is verified and why it implies a statement about words of arbitrary length. The stability theorem of [section 4](#) and the infinite family of [section 7](#) involve no computation over words or labels.

The result compares a source with its complete norm-89 flip. A norm containing other Gaussian prime blocks can have further primitive representations, and no comparison with all of them is asserted here; nor is the critical set (3) claimed to be sharp among genuine Markoff sources. These boundaries leave a concrete question for further work: which recurrent zero-cost components can a central source approach while retaining its arithmetic endpoint conditions?

2. EUCLIDEAN COST AND A COMPLETE GAUSSIAN FLIP

2.1. The cost convention. For a nonzero positive vector $v = (y, x)^T$, let $\ell(v)$ be the number of unit subtractions in the algorithm that repeatedly subtracts the smaller coordinate from the larger and stops when the coordinates are equal. This definition is unchanged by interchanging coordinates or multiplying both by a positive integer. For a vector with nonzero signed coordinates, we apply it to their absolute values.

Write

$$M(j) = \begin{pmatrix} j & 1 \\ 1 & 0 \end{pmatrix}, \quad M(q_1 \cdots q_r) = M(q_1) \cdots M(q_r), \quad e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (5)$$

For a positive continued-fraction word q with final digit at least two, the first column of $M(q)$ is a primitive vector $(y, x)^T$ with $y > x > 0$. The usual canonical finite expansion of y/x is $[q_1; \dots, q_r]$. Division with remainder gives

$$\ell(M(q)e_1) = q_1 + \cdots + q_r - 1. \quad (6)$$

Indeed, the successive divisions make q_i subtractions until the last division, where the algorithm stops at equal coordinates and makes one fewer. The final minus one is part of our convention throughout.

Let χ double each digit, so that $\chi(121) = 112211$. Our regular source language is

$$q = \varepsilon\chi(w)2, \quad \varepsilon \in \{\emptyset, 1, 2\}, \quad w \in \{1, 2\}^*. \quad (7)$$

An empty optional digit contributes zero to a digit sum. If $w = j_1 \cdots j_n$, then

$$\ell(M(q)e_1) = \varepsilon + 2 \sum_{i=1}^n j_i + 1. \quad (8)$$

The language in (7) is larger than the genuine Markoff source language. This enlargement makes the endpoint theorem stronger, while genuine sources will later supply balance and coprime Farey counts.

2.2. The two reflection numerators. The rational prime 89 splits in the ring $\mathbb{Z}[i]$ of Gaussian integers. Fix once and for all the Gaussian prime above it given by

$$\pi = 8 + 5i, \quad \bar{\pi} = 8 - 5i, \quad \pi\bar{\pi} = 89. \quad (9)$$

The four associates $\pm\pi, \pm i\pi$ generate the same ideal, and $\bar{\pi}$ generates the other prime above 89. Every choice made below is expressed through this one Gaussian integer, which makes the construction rigid: the coordinate pair $(8, 5)$ of π is itself the primitive vector of the source $q = 1112$, identified in [corollary 6.2](#) as the unique genuine source of vanishing excess. The two integral matrices

$$R_0 = \begin{pmatrix} 39 & 80 \\ 80 & -39 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 80 & 39 \\ 39 & -80 \end{pmatrix} \quad (10)$$

satisfy $R_\nu^\top R_\nu = 89^2 I$ and $\det R_\nu = -89^2$. Under the identification $(y, x)^\top \leftrightarrow y + ix$, they act as $z \mapsto \pi^2 \bar{z}$ and $z \mapsto i\bar{\pi}^2 \bar{z}$, respectively.

For a nonzero integral vector u , write $\text{cont}(u)$ for the greatest common divisor of its coordinates, taken positive. The next lemma explains why the exact endpoint content, rather than divisibility alone, matters.

Lemma 2.1 (Exact primitive endpoint). *Let $v = (y, x)^\top$ be primitive, with $x, y > 0$, and let $N = x^2 + y^2$. Then $\text{ord}_{89}(N) = 1$ if and only if exactly one of $R_0 v, R_1 v$ has content 89. For this branch, the vector $R_\nu v/89$ is primitive, has nonzero coordinates and norm N . Modulo signs and coordinate order, it represents the complete flip of the Gaussian prime pair above 89.*

Proof. The prime 89 splits as $\pi\bar{\pi}$ in $\mathbb{Z}[i]$. Since v is primitive, $z = y + ix$ cannot be divisible by both π and $\bar{\pi}$. If π^e divides z exactly, with $e > 0$, then the valuations of $\pi^2 \bar{z}$ at $(\pi, \bar{\pi})$ are $(2, e)$. Its rational content at 89 is therefore $89^{\min(2, e)}$. The other branch has content prime to 89. The roles reverse when z is divisible by $\bar{\pi}$. When neither divides z , neither branch has content divisible by 89.

No rational prime other than 89 can divide a reflected content: the matrix is invertible modulo every such prime. Since $N = z\bar{z}$ and only one of $\pi, \bar{\pi}$ divides z , the exponent e is exactly $\text{ord}_{89}(N)$. Thus content exactly 89 is equivalent to $\text{ord}_{89}(N) = 1$. Division gives a primitive norm- N vector. If one coordinate vanished, N would be a square, contradicting $\text{ord}_{89}(N) = 1$.

For example, if $z = \pi t$, the compatible branch divided by 89 is $\pi \bar{t}$. Its conjugate is $\bar{\pi} t$, which exchanges precisely the allocation of π and $\bar{\pi}$. Units and conjugation account for the irrelevant signs and coordinate order. This is the stated complete flip. $\square$

For the branch selected by [lemma 2.1](#), define the completed cost excess

$$\Delta = \ell(R_\nu v/89) - \ell(v). \quad (11)$$

The normalization by 89 does not change the subtractive cost. It does, however, decide whether the terminal vector is the primitive representation being compared. If $\text{ord}_{89}(N) \geq 2$, the compatible raw content is 89^2 and the present theorem does not apply.

3. GENUINE MARKOFF SOURCES AND THEIR BALANCED CORES

The reflection machine will accept continued fractions from a larger language than the one supplied by Markoff occurrences. We now identify the genuine sources inside that language. The distinction matters: balance controls long factors, whereas the exact occurrence also fixes the initial phase and the terminal digit.

3.1. The mechanical word and the Cohn normalization. Fix relatively prime integers $0 < P < Q$. The lower Christoffel word $C_{P/Q} = c_0 \cdots c_{Q-1}$ is defined by

$$c_i = b \iff \left\lfloor \frac{(i+1)P}{Q} \right\rfloor - \left\lfloor \frac{iP}{Q} \right\rfloor = 1. \quad (12)$$

It has P letters b and $Q - P$ letters a ; thus Q is its total length. Its first letter is a and its last is b , so write $C_{P/Q} = azb$. The interior z is a palindrome. Indeed, for $1 \leq j \leq Q - 1$, coprimality gives

$$\left\lfloor \frac{(Q-j)P}{Q} \right\rfloor = P - 1 - \left\lfloor \frac{jP}{Q} \right\rfloor.$$

Applying this identity to two consecutive interior indices shows that $c_i = c_{Q-1-i}$ for $1 \leq i \leq Q - 2$. Such interiors are called central words; see [2] for their standard word-theoretic description.

Put

$$A = M(1)^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = M(2)^2 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad (13)$$

and let μ be the word morphism with $\mu(a) = A$ and $\mu(b) = B$. The Cohn–Christoffel correspondence identifies the normalized Markoff occurrence at P/Q with the label

$$m_{P/Q} = \frac{1}{3} \text{tr } \mu(C_{P/Q}) = \mu(C_{P/Q})_{12}. \quad (14)$$

This is the classical correspondence with Markoff triples, not an assertion that distinct fractions have distinct labels [12]. The two initial Markoff values 1 and 2 lie outside this proper-word normalization and will not be needed below.

3.2. The primitive column and its exact cost. Use the digit product $M(q)$ from (5). Identify a, b with the numerical input letters 1, 2, so that $\chi(a) = 11$ and $\chi(b) = 22$. Reversal is denoted by a tilde.

Proposition 3.1 (The genuine source dictionary). *For every reduced $0 < P < Q$, there is a canonical continued-fraction word*

$$q = \varepsilon \chi(w) 2, \quad \varepsilon \in \{\emptyset, 1, 2\}, \quad (15)$$

whose first column $v = (y, x)^T$ satisfies $y > x > 0$, $\gcd(x, y) = 1$, and $x^2 + y^2 = m_{P/Q}$. The core w is a finite balanced binary word. Moreover,

$$\begin{aligned} P &= |q|_2 = 1 + \mathbf{1}_{\{\varepsilon=2\}} + 2|w|_b, \\ Q &= |q| + 1 = 2 + \mathbf{1}_{\{\varepsilon \neq \emptyset\}} + 2|w|. \end{aligned} \quad (16)$$

and its subtractive Euclidean cost is

$$\ell(v) = \sum_{j=1}^s q_j - 1 = P + Q - 2. \quad (17)$$

The complete word is determined by the central palindrome through

$$\tilde{q}q = 2\chi(z)2. \quad (18)$$

Proof. Split the central palindrome at its middle, in exactly one of the forms

$$z = \tilde{w}w, \quad z = \tilde{w}aw, \quad z = \tilde{w}bw.$$

The corresponding choices are $\varepsilon = \emptyset, 1, 2$. Since A and B are symmetric, in all three cases

$$\mu(z) = H^\top H, \quad H = \begin{cases} \mu(w), & \varepsilon = \emptyset, \\ M(\varepsilon)\mu(w), & \varepsilon \in \{1, 2\}. \end{cases} \quad (19)$$

Let $e = (2, 1)^\top = M(2)(1, 0)^\top$. The first row of A is $e^\top$ and the second column of B is e . Consequently

$$\mu(azb)_{12} = e^\top \mu(z) e = \|He\|^2.$$

For completeness, if $\mu(z) = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, direct multiplication gives

$$\mu(azb)_{12} = 4a + 4b + d, \quad \text{tr } \mu(azb) = 3(4a + 4b + d),$$

which also verifies the equality of the two label conventions in (14).

The matrix H is integral and unimodular. Therefore $v = He$ is primitive. Its coordinates are positive, with the first larger than the second: every digit matrix $M(j)$ sends a positive column to one with this ordering. By construction $HM(2) = M(q)$, proving the norm claim. The last digit of q is 2, so q is the canonical finite continued fraction of y/x . This also includes the empty center, where $q = 2$ and $v = (2, 1)^\top$.

Reversing the digit word in each of the three middle cases gives (18) literally. In particular, the full word has length $2 + 2|z| = 2Q - 2$ and contains $2 + 2|z|_b = 2P$ digits equal to 2. Its two reversed halves have equal length and equal digit counts. Hence $|q| = Q - 1$ and $|q|_2 = P$, which gives (16).

Equation (6) gives $\ell(v) = \sum q_j - 1$. Since all digits are 1 or 2, their sum is $|q| + |q|_2 = Q - 1 + P$, proving (17).

Finally, a binary word is *balanced* if any two of its contiguous factors of equal length have numbers of b 's differing by at most one. For a length- r factor of $C_{P/Q}$ starting at i , the number of b 's is, by telescoping (12),

$$\left\lfloor \frac{(i+r)P}{Q} \right\rfloor - \left\lfloor \frac{iP}{Q} \right\rfloor.$$

For fixed r this is either $\lfloor rP/Q \rfloor$ or $\lceil rP/Q \rceil$. Thus $C_{P/Q}$ is balanced, as is every contiguous factor. The right half w of z is one such factor. $\square$

The proof explains both roles of the source grammar. Pairing the digits produces an integer Gram square and an exact cost; the mechanical word supplies balance. No cyclic balance of an arbitrary central half is asserted. In particular, a repeated segment of a finite core must be analyzed with its actual initial and terminal states.

Conversely, the conditions that q has the form (15), that w is balanced, and that the counts in (16) are coprime are only necessary conditions in this paper. An exact recognition test must also compare $\tilde{q}q$ with $2\chi(z_{P/Q})2$, where $z_{P/Q}$ is the interior computed from (12). If this equality holds, the first-column construction proves that the source is exactly the occurrence at P/Q . The reflection arguments below apply to a larger language when this final recognition test is not imposed.

3.3. How thin the genuine language is. Balance alone is a weak constraint compared with being a genuine core. The classical enumeration of the finite factors of Sturmian words makes the comparison exact. Those factors are precisely the balanced words, and there are $1 + \sum_{i=1}^n (n-i+1)\varphi(i)$ of length n , with φ the Euler function; Mignosi proved this, and Berstel and Pocchiola gave a geometric proof by duality and Euler's relation for planar graphs [8, 1].

Proposition 3.2 (The genuine cores of a given length). *Fix $n \geq 1$. The pairs (ε, w) produced by proposition 3.1 with $|w| = n$ are in bijection with the reduced fractions P/Q , $0 < P < Q$, having $Q \in \{2n+2, 2n+3\}$. There are therefore exactly*

$$\varphi(2n+2) + \varphi(2n+3) \leq 3n+3 \quad (20)$$

of them, whereas the balanced words of length n number $1 + \sum_{i=1}^n (n-i+1)\varphi(i)$, which is asymptotic to n^3/π^2 .

Proof. By (16) one has $Q = 2 + \mathbf{1}_{\{\varepsilon \neq \emptyset\}} + 2n$, so $|w| = n$ forces $Q = 2n+2$ when $\varepsilon = \emptyset$ and $Q = 2n+3$ otherwise; conversely both values occur, the parity of Q being the parity of $|z| = Q - 2$ that decides the three middle cases in the proof of proposition 3.1. That proof assigns to each reduced P/Q exactly one pair, so the map is well defined. It is injective because the pair recovers the central word, $z = \tilde{w}w$, $\tilde{w}aw$ or $\tilde{w}bw$ according as ε is $\emptyset$, 1 or 2, and distinct reduced fractions have distinct central words. Counting reduced fractions with a prescribed denominator gives (20); the bound follows from $\varphi(2n+2) \leq n+1$, valid because $2n+2$ is even, together with $\varphi(2n+3) \leq 2n+2$. Finally $\sum_{i \leq x} \varphi(i) \sim 3x^2/\pi^2$, and summing this once more over $i \leq n$ gives the stated asymptotic. $\square$

At length n the reflection machine may therefore be fed any of about n^3/π^2 balanced words, of which at most $3n+3$ come from a genuine occurrence: a proportion $O(n^{-2})$. This is the precise sense in which the arithmetic language is narrower than the combinatorial one, and it locates where a sharper form of the results below would have to come from, since balance is blind to the distinction.

A second consequence of genuineness is available at once, and it removes an auxiliary choice from the argument of section 7: the Farey slope of an occurrence may be used directly as a compatible slope for its core, with no recourse to the existence statement of lemma 4.1.

Corollary 3.3 (The Farey slope is compatible). *Let w be the core of a genuine reduced occurrence of slope P/Q . Then every nonempty factor v of w satisfies*

$$||v|_b - (P/Q)|v|| < 1. \quad (21)$$

Proof. The word w is a factor of $C_{P/Q}$, so by the telescoping computation at the end of the proof of proposition 3.1 a factor v of length r has $|v|_b$ equal to $\lfloor rP/Q \rfloor$ or $\lceil rP/Q \rceil$. Moreover $r \leq |w| < Q$ and $\gcd(P, Q) = 1$, so rP/Q is not an integer; each of the two values is therefore at distance strictly less than one from it. $\square$

4. BALANCED FINITE PATHS NEAR CRITICAL FREQUENCIES

This section is independent of Gaussian arithmetic. It isolates the mechanism that turns a finite graph with nonnegative costs into a quantitative statement about the words of small cost it accepts.

4.1. One slope controls every factor. A finite binary word is *balanced* if any two of its factors of the same length have heavy-letter counts differing by at most one. We denote the heavy letter by b in this section. The following classical form of finite balance is useful because the same slope works for every factor.

Lemma 4.1 (A compatible slope). *For every nonempty finite balanced word w , there is an irrational $\beta \in (0, 1)$ such that every factor v satisfies*

$$||v|_b - \beta|v|| < 1. \quad (22)$$

Proof. For two nonempty factors u, v , put $p = |u|$, $q = |v|$, $A = |u|_b$ and $B = |v|_b$. We first prove

$$|qA - pB| < p + q \quad (23)$$

by induction on $p + q$. If $p = q$, balance bounds the left side by p . Otherwise interchange the factors so that $p < q$, and write $v = v_1v_2$ with $|v_1| = p$. Then

$$qA - pB = p(A - |v_1|_b) + (q - p)A - p|v_2|_b.$$

Balance bounds the first term in absolute value by p , and the induction hypothesis bounds the remaining two terms together strictly by q . This proves (23), which after division by pq is the factorial-balance inequality of the classical theory of Sturmian words [7, Proposition 2.1.7].

For each nonempty factor v , take the open interval with endpoints $(|v|_b - 1)/|v|$ and $(|v|_b + 1)/|v|$. Inequality (23) says that every two intervals overlap. There are finitely many intervals, so their largest lower endpoint is strictly smaller than their smallest upper endpoint. Every lower endpoint is below one and every upper endpoint is positive. The common intersection with $(0, 1)$ is therefore nonempty and open. Any irrational β in this intersection proves (22). $\square$

The passage from that inequality to a single compatible slope is also classical; see [7, Chapter 2]. We included the proof to keep the finite-path argument self-contained. The compatible slope need not equal the empirical density $|w|_b/|w|$. For instance, 00100010000 is balanced, but its factor 10001 has discrepancy 12/11 relative to that density. The common slope in lemma 4.1 avoids this finite-word endpoint error.

4.2. The slope interval as a projection. The interval constructed in the proof of lemma 4.1 is not an artefact of that proof. It is the shadow, on the slope axis, of a planar region attached to w , and the discrepancy condition is exactly what survives when the second parameter is eliminated.

Read b as 1 and a as 0, and recall that the *mechanical word* with slope β and intercept γ has i th letter $\lfloor \beta i + \gamma \rfloor - \lfloor \beta(i-1) + \gamma \rfloor$.

Proposition 4.2 (The compatible slopes are a projection). *Let $w = w_1 \cdots w_n$ be a nonempty binary word, put $k_i = |w_1 \cdots w_i|_b$ for $0 \leq i \leq n$, and set*

$$\mathcal{F}(w) = \{(\beta, \gamma) \in \mathbb{R}^2 : k_i \leq \beta i + \gamma < k_i + 1, \quad 0 \leq i \leq n\}. \quad (24)$$

Then $\mathcal{F}(w)$ is the set of parameters whose mechanical word of length n is w , and its image under $(\beta, \gamma) \mapsto \beta$ is

$$J(w) = \{\beta \in \mathbb{R} : ||v|_b - \beta|v|| < 1 \text{ for every nonempty factor } v \text{ of } w\}, \quad (25)$$

an open interval with rational endpoints. It is nonempty precisely when w is balanced, and the slope of lemma 4.1 is any irrational point of $J(w) \cap (0, 1)$.

Proof. The constraint at $i = 0$ reads $0 \leq \gamma < 1$, so on $\mathcal{F}(w)$ one has $\lfloor \beta i + \gamma \rfloor = k_i$ for every i , and the successive differences of the k_i are the letters of w ; conversely those differences determine the k_i from $k_0 = 0$. This is the first assertion.

Eliminate γ . The system (24) has a solution γ for a given β exactly when

$$\max_{0 \leq i \leq n} (k_i - \beta i) < \min_{0 \leq j \leq n} (k_j + 1 - \beta j),$$

that is, when $(k_i - k_j) - \beta(i - j) < 1$ for all $i \neq j$, the case $i = j$ being vacuous. For $i > j$ the factor $v = w_{j+1} \cdots w_i$ has $|v|_b = k_i - k_j$ and $|v| = i - j$, so the condition is $|v|_b - \beta|v| < 1$; for $i < j$ the same factor, taken with the roles reversed, gives $\beta|v| - |v|_b < 1$. The two families together are (25). Being a finite intersection of the open intervals with endpoints $(|v|_b \mp 1)/|v|$, the set $J(w)$ is an open interval with rational endpoints.

If $\beta \in J(w)$ and u, v are factors of a common length p , then $||u|_b - |v|_b| < 2$, hence at most one by integrality, so w is balanced; the converse is lemma 4.1, whose proof exhibits a point of $J(w)$ in $(0, 1)$. An open interval contains an irrational point. $\square$

Remark 4.3 (The two-parameter picture). The regions $\mathcal{F}(w)$, for w ranging over the balanced words of length n , are the faces of an arrangement of lines in the (β, γ) plane: each constraint in (24) is a half-plane, and the bounding lines are dual to the lattice points involved. Berstel and Pocchiola used exactly this duality, together with Euler's relation, to count the faces and so to recover Mignosi's enumeration of the finite factors of Sturmian words, of which there are $1 + \sum_{i=1}^n (n - i + 1)\varphi(i)$ of length n , with φ the Euler function [1, 8]. In that reading, proposition 4.2 says that lemma 4.1 computes the shadow of one face, and that the intercept is the coordinate being integrated out. The two parameters play complementary roles: over a fixed irrational slope the fibre of the projection is cut into $n + 1$ cells, one for each factor of that length of a Sturmian word of that slope, whereas moving the slope changes the face, and so the word itself.

4.3. Endpoint weights and critical components. Let G be a finite directed graph. Each edge reads exactly one letter in $\{a, b\}$; parallel edges and nondeterminism are allowed. Initial states have weights $I(s)$, terminal states have weights $T(s)$, and edges have weights $c(e)$. Assume all these weights are nonnegative integers. The completed weight of an accepted path is

$$\Delta = I(s_0) + \sum_{e \text{ on the path}} c(e) + T(s_n). \quad (26)$$

These weights may result from a potential transformation. If original weights are $w_0, w(e), f$ and a potential is λ , the relevant transformation is

$$I(s_0) = w_0 - \lambda(s_0), \quad c(e) = w(e) + \lambda(s) - \lambda(t), \quad T(s_n) = \lambda(s_n) + f(s_n). \quad (27)$$

This is the usual weight-pushing transformation [9, Section 6.3]. The terms telescope, preserving the completed cost. Nonnegative cycle weights alone do not guarantee nonnegative values of both transformed endpoint weights.

Let G_0 retain the edges of weight zero. Its strongly connected components form an acyclic condensation graph $\mathcal{D}$. Call a component *cyclic* if it contains a directed cycle, including a self-loop. Suppose each cyclic component C_i admits a reduced rational number $\theta_i = p_i/d_i$, with $d_i > 0$, and an integer function H_i on its vertices such that

$$d_i \mathbf{1}_{\{e \text{ reads } b\}} - p_i = H_i(t) - H_i(s) \quad (28)$$

for every internal edge $e : s \rightarrow t$. Put

$$D_i = \frac{\max H_i - \min H_i}{d_i}, \quad \Theta = \{\theta_i : C_i \text{ cyclic}\}. \quad (29)$$

Summation around a cycle gives $0 \leq \theta_i \leq 1$. Summation along an internal path gives

$$||v|_b - \theta_i|v|| \leq D_i. \quad (30)$$

Condition (28) is equivalent to all directed cycles in the component having the same letter frequency. To see the converse, assign to a vertex the sum of $d_i \mathbf{1}_b - p_i$ along a

path from a fixed root. Two choices give the same sum: append a common return path, then decompose the resulting closed walks into cycles. This also shows how to check the condition by a finite list of edge identities.

Give each cyclic vertex of $\mathcal{D}$ weight $D_i + 1$, every other vertex weight zero and every condensation edge weight one. Define

$$L = \max_{\gamma \text{ a directed path in } \mathcal{D}} \left(\# \text{edges of } \gamma + \sum_{C_i \in \gamma \text{ cyclic}} (D_i + 1) \right), \quad K = L + 1. \quad (31)$$

Paths consisting of a single vertex are included. The number L is finite and is obtained by the usual longest-path recursion on an acyclic graph.

It is convenient to allow a bounded number of letters to be read outside the graph. Fix an integer $h \geq 0$, and suppose that an initial prefix and a final suffix of total length at most h are consumed before entering and after leaving G , while (26) still records the completed cost of the whole word. The case in which no letter is omitted is $h = 0$.

4.4. The stability theorem. We isolate first the counting step, because it uses only one property of a real number β , namely that it be compatible with w in the sense of (22), and not the particular slope furnished by lemma 4.1. The arithmetic application in section 7 will supply its own compatible slope.

Lemma 4.4 (The budget inequality). *Assume $\Theta \neq \emptyset$. Let w be a nonempty word of length n , all but at most h of whose letters are read by an accepted path, the remaining ones forming a prefix and a suffix consumed outside the graph. Let β be a real number with $||v|_b - \beta|v|| < 1$ for every nonempty factor v of w , and put $\delta = \text{dist}(\beta, \Theta)$. If $\delta \leq 1$, then*

$$n\delta \leq h + \Delta + (\Delta + 1)L. \quad (32)$$

Proof. Let k be the number of positive-weight edges in the path. Integrality and nonnegative endpoint weights imply $k \leq \Delta$.

Split the path into k positive edges and $k + 1$ possibly empty zero-edge segments. A zero segment visits each condensation vertex at most once. Inside a cyclic component, its internal subpath reads a factor v of the full word, and the hypothesis on β together with (30) implies

$$\delta|v| \leq |\beta - \theta_i||v| < 1 + D_i.$$

A noncyclic component is a singleton without an internal edge. Every remaining letter crosses a condensation edge and contributes at most one to δ times length, since $\delta \leq 1$. Thus each zero segment contributes at most L . The positive edges contribute at most k , and the omitted boundary letters at most h . Consequently $n\delta \leq h + k + (k + 1)L$, which is (32) because $k \leq \Delta$ and $L \geq 0$. $\square$

Theorem 4.5 (Balanced finite-path stability). *Assume $\Theta \neq \emptyset$. Let w be a balanced word all but at most h of whose letters are read by an accepted path, the remaining ones forming a prefix and a suffix consumed outside the graph. Then*

$$\min_{\theta \in \Theta} ||w|_b - \theta|w|| < K(\Delta + 1) + h. \quad (33)$$

Both counts on the left refer to the whole word. In particular, when no letter is omitted the bound is $K(\Delta + 1)$.

Proof. The empty word is immediate. Otherwise put $n = |w|$, $m = |w|_b$, and choose β from lemma 4.1; it satisfies (22), and $\delta = \text{dist}(\beta, \Theta) \leq 1$ because β and every member of Θ lie in $[0, 1]$. Lemma 4.4 applies. Choosing a member of Θ nearest to β and applying (22) also to the whole word gives

$$\min_{\theta \in \Theta} |m - \theta n| < 1 + n\delta \leq h + (\Delta + 1)(L + 1) \leq h + K(\Delta + 1).$$

□

Corollary 4.6 (Linear cost away from the critical set). *Under the hypotheses of theorem 4.5, if $n > 0$ and $\text{dist}(m/n, \Theta) \geq \eta > 0$, then*

$$\Delta > \frac{\eta n - h}{K} - 1.$$

Consequently, for a fixed h , every accumulation point of the empirical frequencies of balanced accepted words of bounded completed cost and unbounded length lies in Θ .

If G_0 has no cyclic component, one must not take a minimum over an empty set. In this case each zero segment has at most $c - 1$ edges, where c is the number of components. The corresponding conclusion is the direct length bound $n \leq h + c(\Delta + 1) - 1$.

4.5. Why the finite set is stronger than its convex hull. Assume again that $\Theta \neq \emptyset$. Without balance, let

$$L_0 = \max_{\gamma \subset \mathcal{D}} \left(\# \text{edges of } \gamma + \sum_{C_i \in \gamma \text{ cyclic}} D_i \right).$$

Assign frequency θ_i to every internal cyclic-component segment, and any fixed member of Θ to the remaining letters. The length-weighted average of these assigned frequencies belongs to $\text{conv } \Theta$. Telescoping the internal count potentials therefore gives, for an arbitrary nonempty input,

$$n \text{dist}(m/n, \text{conv } \Theta) \leq h + k + (k + 1)L_0. \quad (34)$$

If all critical frequencies coincide, this already controls the count without balance.

In general the finite set cannot replace its convex hull. Take a zero-cost a loop at an initial state, a zero-cost b loop at a terminal state, and a zero-cost a edge between them. Both endpoint weights are zero. The critical set is $\{0, 1\}$, yet the accepted words $a^{r+1}b^r$ have cost zero and count distance r from that set. For $r \geq 2$ these words are unbalanced. The example identifies exactly the obstruction removed by lemma 4.1.

5. THE ENDPOINT INEQUALITY AT NORM 89

Theorem 5.1 (The norm-89 endpoint inequality). *Let $q = \varepsilon\chi(w)2$ be any word in (7), put $v = M(q)e_1$, and assume $\text{ord}_{89}(\|v\|^2) = 1$. Then its complete norm-89 flip satisfies $\Delta \geq 0$.*

The proof replaces the infinitely many admissible words by the finitely many states of a reduction graph, and then exhibits on that graph an integer potential making every step of the cost comparison nonnegative. The finiteness of the graph is not an empirical stopping observation: it is forced by the determinant.

5.1. From row subtraction to weighted transitions. Suppose a nonnegative integral matrix S has two incomparable rows: neither dominates the other componentwise. After multiplying by $M(j)^2$, subtract a dominated row from the dominating row until the rows are again incomparable. These are the elementary row operations

$$(a, b; c, d) \mapsto (a - c, b - d; c, d) \quad \text{or} \quad (a, b; c, d) \mapsto (a, b; c - a, d - b).$$

The notation lists the two rows separated by a semicolon. Every such subtraction is forced for *every* positive continuation vector: componentwise domination implies domination after multiplication by that vector. The rows remain nonnegative, and the sum of the entries decreases, so this reduction terminates. If k unit subtractions are emitted, assign the transition the weight

$$w(S, j) = k - 2j. \quad (35)$$

The subtracted $2j$ is exactly the source cost of the paired digit jj .

Row exchange has no effect on Euclidean cost. We therefore identify a matrix with its row exchange and choose the representative with positive determinant. Every reduced representative takes the form

$$S = \begin{pmatrix} r+u & b \\ r & b+v \end{pmatrix}, \quad r, b \geq 0, \quad u, v \geq 1. \quad (36)$$

Lemma 5.2 (State invariants). *Every state reachable from a seed has the crossed form (36), content one, and determinant*

$$rv + bu + uv = 89^2. \quad (37)$$

In particular $r, b, u, v \leq 89^2$, so only finitely many states occur.

Proof. A seed arises from $R_\nu M(\varepsilon)$ by reading paired letters and performing row operations. The digit matrices are unimodular and $\det R_\nu = -89^2$, so a seed has determinant $\pm 89^2$; the choice of the representative with positive determinant fixes the sign, and expanding (36) gives (37). Row subtraction, row exchange and sign change are unimodular, and so are the digit matrices acting on the right; each therefore preserves both the determinant up to sign and the greatest common divisor of the entries. Since a seed has content one, so does every state reachable from it. Finally (37) writes 89^2 as a sum of three nonnegative products, and $u, v \geq 1$, so each of r, b, u, v is at most 89^2 . $\square$

If the input ends in state S , its remaining vector is Se , where $e = (2, 1)^\top$. The terminal charge is

$$f(S) = \ell(Se) - 1. \quad (38)$$

We accept the terminal exactly when $\text{cont}(Se) = 89$. Unimodular row operations preserve vector content, so lemma 2.1 makes this the exact primitive endpoint test. The final one subtracted in (38) is the source cost of its last digit 2.

5.2. Resolving every initial sign. The first row of each R_ν in (10) is positive, but the second has opposite signs. Starting with $R_\nu M(\varepsilon)$, read paired letters until the second row has one weak sign componentwise. At that point its sign is fixed for every positive continuation. Change its sign if necessary, perform the forced row reduction, and enter the finite graph.

This prefix procedure has eleven leaves. Table 1 gives every leaf state and its seed weight. A seed that consumes the paired prefix p and emits k_0 subtractions has weight

$$w_0 = k_0 - \varepsilon - 2 \sum_{j \in p} j. \quad (39)$$

The optional digit is again interpreted as zero when absent.

The prefix tree has sixteen nodes, so besides the eleven leaves there are five words that stop earlier. They are the empty paired prefix with absent optional digit in each branch, the empty prefix with optional digit 1 or 2 in branch R_1 , and prefix 2 with absent optional digit in branch R_1 . Appending the final digit 2 gives raw contents 1, 1, 1, 1, 1, respectively, so none is accepted. The seed table therefore covers every primitive source in the theorem, including short words, and no infinite sign-resolution branch is left over.

5.3. The potential and the proof. Starting from the eleven seeds, close the state set under the two transitions. The next statement describes the resulting graph. It is a finite assertion about integers, and it is the one place in this section where we appeal to a computation.

TABLE 1. Complete sign-resolution seeds. A state $(a, b; c, d)$ has the displayed first and second rows and positive determinant 89^2 . The prefix consists of paired input letters. These eleven leaves cover every continuing word after at most two such letters.

Branch	ε	Prefix	State S_0	w_0
R_0	$\emptyset$	1	(121, 41; 37, 78)	-1
R_0	$\emptyset$	2	(223, 10; 33, 37)	0
R_0	1	$\emptyset$	(119, 39; 41, 80)	-1
R_0	2	$\emptyset$	(158, 39; 121, 80)	-2
R_1	$\emptyset$	1	(195, 37; 2, 41)	0
R_1	$\emptyset$	21	(68, 33; 67, 149)	10
R_1	$\emptyset$	22	(119, 1; 52, 67)	9
R_1	1	1	(43, 2; 17, 185)	4
R_1	1	2	(127, 43; 120, 103)	0
R_1	2	1	(233, 20; 35, 37)	3
R_1	2	2	(271, 23; 68, 35)	7

Proposition 5.3 (The closed graph and its potential). *The closure of the eleven seeds of table 1 under the two transitions has 11420 states and 22840 labeled edges, of which exactly 116 states satisfy the primitive terminal condition. Let $\lambda(S)$ denote the least weighted length of a path from a seed to S , each seed contributing its own weight w_0 . Then*

$$\begin{aligned}
w_0 - \lambda(S_0) &\geq 0 && \text{at all eleven seeds,} \\
w(S, j) + \lambda(S) - \lambda(S') &\geq 0 && \text{on all 22840 edges,} \\
\lambda(S) + f(S) &\geq 0 && \text{at all 116 primitive terminals.}
\end{aligned} \tag{40}$$

Proof. By lemma 5.2 the state set is finite, and every quantity appearing in (40) is an integer; the assertion is therefore a finite list of integer inequalities. We have verified it by computer, in exact integer arithmetic: the state set is closed under the two transitions, the minimum defining d is attained and computed at every state, and the three families are then checked on the closed graph. The verification was carried out twice, independently, and the resulting table of values of λ accompanies this article as supplementary material. $\square$

The values of λ may be negative; it is their transformed combinations in (40) that are nonnegative.

Proof of theorem 5.1. Let q be an accepted word, with seed state S_0 and terminal state S_f . Every operation emitted along its path is an actual unit subtraction on the reflected continuation: the two entries of Sc can never be equal at an emitted step, since the dominating row minus the dominated one would then annihilate the positive continuation vector c , making a row of S vanish against c and forcing $\det S = 0$, against (37); the unconsumed terminal cost is (38), and the consumed source charges are exactly (39) and (35). Consequently

$$\begin{aligned}
\Delta &= w_0 + \sum_e w(e) + f(S_f) \\
&= (w_0 - \lambda(S_0)) + \sum_{e: S \rightarrow S'} (w(e) + \lambda(S) - \lambda(S')) + \lambda(S_f) + f(S_f),
\end{aligned} \tag{41}$$

the inserted potentials telescoping along the path. Every term on the second line is nonnegative by [proposition 5.3](#). No bound on the length of q entered the argument, so the inequality holds for arbitrary finite input length. $\square$

Remark 5.4. It is worth separating the two contributions. The mathematics lies in [lemma 5.2](#) and in the telescoping identity (41): the first makes the state set finite, the second turns three families of inequalities on that finite set into a statement about words of unbounded length. The computation contributes only the closed graph and the integer potential, and both are recomputed from the seeds rather than read from a stored table.

6. THE COMPLETE EQUALITY LANGUAGE

The potential proves more than nonnegativity. Equality in (41) forces every transformed weight along the path to vanish, and this reduces an inequality over 11420 states to a language on six of them.

Theorem 6.1 (Equality and the reflected word). *Under the hypotheses of [theorem 5.1](#), equality $\Delta = 0$ holds exactly when*

$$q = 111\chi(s)2, \quad s \in (12111 \mid 21111)^*.$$

The target is $111\chi(\iota(s))2$, where ι exchanges the two five-letter tokens, preserving their order in the token sequence.

6.1. Exhausting the zero-slack paths. Retain every edge whose transformed weight is zero. In this graph, keep the states that are both reachable from a zero-slack seed and able to reach a zero-slack primitive terminal. The complete calculation gives exactly the states in [table 2](#). The only surviving seed is branch R_0 , optional digit 1, with empty paired prefix and state s_0 . The only surviving terminal is s_1 .

TABLE 2. The full equality graph. The last column is the terminal charge $f(S)$ when the state is a primitive terminal; a dash means the state is not such a terminal. Only s_0 is an initial state and only s_1 is accepting.

State	Matrix S	$\lambda(S)$	$f(S)$
s_0	(119, 39; 41, 80)	−1	−
s_1	(115, 37; 47, 84)	−1	1
s_2	(89, 21; 0, 89)	0	−
s_3	(89, 0; 68, 89)	−1	−
s_4	(131, 21; 47, 68)	−1	−
s_5	(121, 37; 41, 78)	−1	−

Every edge is shown in [figure 1](#). From s_0 , the first letter must be 1, reaching s_1 . The two first-return words at s_1 are exactly 12111 and 21111. Thus the accepted paired words are $1(12111 \mid 21111)^*$. The optional initial digit 1 converts this to the asserted source language, which proves necessity.

That the returns at s_1 are two in number, and not one or three, is what makes the equality language a free monoid on two generators, and hence what makes the count relation $Q = 5P$ of [corollary 6.2](#) rigid. The same numerical signature characterizes Sturmian words, whose every factor has exactly two return words [13, 6]. The two notions are distinct: returns to a factor of an infinite word on one side, first returns to a state of a weighted graph on the other. We assert no implication between them. Both tokens above nevertheless carry one heavy letter in five, which is the frequency $\theta = 1/5$ that these same six states contribute to the critical set of [section 7](#).

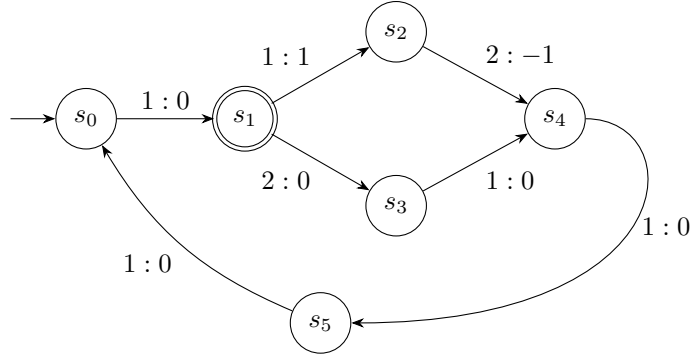


FIGURE 1. All zero-slack accepted paths at norm 89. An arrow label $j : w$ gives the paired input letter and its original edge weight; the transformed weight is zero on every displayed arrow. The seed is s_0 and the double circle marks the only accepting state s_1 . Its two first-return words are 12111 and 21111.

6.2. An exact token exchange. The literal reflected word follows from small matrix identities. Put

$$Q_0 = M(1)^3, \quad K = \begin{pmatrix} 47 & 84 \\ 68 & -47 \end{pmatrix}, \quad H = \begin{pmatrix} 89 & 68 \\ 0 & -89 \end{pmatrix}. \quad (42)$$

Direct multiplication gives

$$R_0 Q_0 = Q_0 K, \quad K M(2) = M(2) H. \quad (43)$$

Writing $G_t = M(\chi(t))$, the two return identities are

$$K G_{12111} = G_{21111} K, \quad K G_{21111} = G_{12111} K. \quad (44)$$

Induction over the token sequence therefore yields

$$R_0 M(111\chi(s)2) = M(111\chi(\iota(s))2)H. \quad (45)$$

The first column of H is $89e_1$. Hence the reflected first column is 89 times the primitive positive continuant column of the displayed target. The source and target have equal digit sums. This proves sufficiency and the target formula in [theorem 6.1](#), including exact primitive acceptance, for every token sequence.

Corollary 6.2 (Strict cost increase for genuine sources). *Let v be the canonical source of a reduced Markoff occurrence, with $\text{ord}_{89}(\|v\|^2) = 1$. Every distinct complete norm-89 target has cost at least $\ell(v) + 1$. The only genuine equality source is $q = 1112$, with $v = (8, 5)^\top$, which the flip fixes.*

Proof. If an equality word contains j five-letter tokens, its inferred counts are

$$P = 1 + 2j, \quad Q = 5 + 10j = 5P.$$

A genuine reduced occurrence has $\gcd(P, Q) = 1$ by definition, and [proposition 3.1](#) identifies the counts of its word with that P and Q . Therefore $j = 0$. The base column is $(8, 5)^\top$ and [\(45\)](#) fixes it after division by 89. All other genuine sources have positive integral excess. $\square$

The proof uses coprime counts only after exhausting the equality language. It does not use the converse assertion that a balanced paired word with coprime counts is automatically a genuine occurrence.

TABLE 3. Every cyclic zero-slack component at norm 89. The frequency θ_i counts paired letters equal to 2. The oscillation D_i is that of the rational count potential $g_i = H_i/d_i$. Every internal edge satisfies $\mathbf{1}_{\{j=2\}} - \theta_i = g_i(S') - g_i(S)$.

Number of components	States per component	θ_i	D_i
1	6	1/5	1
2	3	1/3	2/3
2	1	0	0
2	1	1	0

7. QUANTITATIVE PENALTIES FOR GENUINE MARKOFF SOURCES

The equality graph describes paths with total excess zero. Near-minimizers can also visit zero-slack components that have a positive cost of entry or exit. To understand them, we must examine the entire zero-slack graph, not only the six states surviving the equality test.

7.1. The full critical set. We first record the structure of the zero-slack subgraph.

Proposition 7.1 (The critical components at norm 89). *The zero-slack subgraph of (40) has 12684 edges and 11411 strongly connected components, of which exactly seven are cyclic. Their frequencies and oscillations are those listed in table 3, and every internal edge of each satisfies the count-potential identity (28). The longest path in the condensation, weighted as in (31), has*

$$L = 58, \quad K = 59. \quad (46)$$

Proof. All of this concerns the finite graph of proposition 5.3, so it is again a finite assertion about integers, and we verified it by computer in exact arithmetic, twice and independently. A path realizing the maximum traverses 56 condensation edges together with the six-state component, whose vertex weight is two; this accounts for the value of L . The states and internal edges of the seven cyclic components are listed in the supplementary material. $\square$

Theorem 7.2 (Norm-89 balanced stability). *Under the hypotheses of theorem 5.1, suppose the core w is balanced. With $n = |w|$, $m = |w|_2$ and $\Theta_{89} = \{0, 1/5, 1/3, 1\}$,*

$$\min_{\theta \in \Theta_{89}} |m - \theta n| < 59(\Delta + 1) + 2. \quad (47)$$

If the source is a genuine reduced occurrence of slope P/Q , then

$$\min\{P, Q - P, |P - Q/5|, |P - Q/3|\} \leq 118\Delta + 121. \quad (48)$$

Proof. Proposition 5.3 supplies the nonnegative integer weights required by theorem 4.5, and proposition 7.1 gives $L = 58$ and $K = 59$. At most two paired letters are consumed in the sign-resolution prefix, so the theorem with $h = 2$ gives (47).

For a genuine occurrence the auxiliary slope can be dispensed with. By corollary 3.3 the Farey slope P/Q is itself compatible with the core, so lemma 4.4 may be applied to $\beta = P/Q$. Its hypothesis $\delta \leq 1$ holds with room to spare: the complementary intervals of Θ_{89} in $[0, 1]$ are $[0, 1/5]$, $[1/5, 1/3]$ and $[1/3, 1]$, whose half-lengths are $1/10$, $1/15$ and $1/3$, so

$$\delta = \text{dist}(P/Q, \Theta_{89}) \leq \frac{1}{3}. \quad (49)$$

The left side of (48) is exactly $Q\delta$. Writing $\mathbf{1}_\neq$ for the indicator of $\varepsilon \neq \emptyset$, the count dictionary (16) gives $Q = 2n + 2 + \mathbf{1}_\neq$, whence

$$Q\delta = 2n\delta + (2 + \mathbf{1}_\neq)\delta \leq 2(h + \Delta + (\Delta + 1)L) + 3\delta.$$

With $h = 2$, $L = 58$ and (49) the right side is $2(2 + \Delta + 58(\Delta + 1)) + 1 = 2(59\Delta + 60) + 1 = 118\Delta + 121$. $\square$

The gain over the route through (47), which would give $118\Delta + 124$, is small. What matters is that no slope other than the occurrence's own enters the argument: [corollary 3.3](#) replaces an existence statement about balanced words by an identity about the Farey address, and this is the only place where the narrowness recorded in [proposition 3.2](#) is actually used.

Corollary 7.3 (Slope-dependent linear penalty). *If a genuine source in [theorem 7.2](#) satisfies $\text{dist}(P/Q, \Theta_{89}) \geq \eta > 0$, then*

$$\Delta \geq \frac{\eta Q - 121}{118}.$$

Thus any sequence of genuine sources with bounded norm-89 excess and unbounded denominators can accumulate only at the four critical slopes.

Proof. For each $\theta \in \Theta_{89}$ one has $|P - \theta Q| = Q |P/Q - \theta| \geq \eta Q$, so the left side of (48) is at least ηQ ; rearranging that inequality gives the bound. The last assertion is the contrapositive: along a sequence with Δ bounded and $Q \rightarrow \infty$ the quantity $(\eta Q - 121)/118$ is eventually larger than any bound on Δ , so no positive η can persist. $\square$

The constants arise from the full condensation and were not optimized using the genuine source language. In particular, the last assertion is a necessary condition on accumulation points, not an existence theorem at each critical slope.

7.2. An infinite arithmetic family with growing excess.

Theorem 7.4 (A genuine infinite family). *There is an infinite sequence of genuine reduced Markoff occurrences P_j/Q_j whose labels have exact 89-adic valuation one, for which $P_j, Q_j \rightarrow \infty$ and*

$$\frac{P_j}{Q_j} \rightarrow \rho = \frac{1 + \Phi}{4 + 5\Phi}, \quad \Phi = \frac{1 + \sqrt{5}}{2} \quad \text{the golden ratio.} \quad (50)$$

For all sufficiently large j , their complete norm-89 excesses satisfy

$$\Delta_j \geq \frac{Q_j}{1180(4 + 5\Phi)} - \frac{121}{118}. \quad (51)$$

Proof. Start with the Markoff triple $(1, 34, 89)$ and iterate

$$F(a, b, c) = (b, c, 3bc - a). \quad (52)$$

The identity $a^2 + b^2 + c^2 = 3abc$ is preserved. If $1 \leq a < b < c$, then $3bc - a > c$, so the positive triples continue forward in the Markoff tree and their maxima increase strictly.

Modulo 89^2 , the same map is a permutation because it has the polynomial inverse

$$F^{-1}(a, b, c) = (3ab - c, a, b).$$

The orbit of the initial residue triple is therefore periodic from its first term. Let $T > 0$ be a period. For every $j \geq 0$, the last coordinate of $F^{jT}(1, 34, 89)$ is congruent to 89 modulo 89^2 . It consequently has valuation exactly one at 89.

The initial labels 34 and 89 occur at the adjacent Farey vertices $1/4$ and $1/5$; their other common neighbour $0/1$ has label one. The Vieta step is exactly the Farey mediant: the Christoffel word of the mediant of two Farey neighbours is the concatenation of their

Christoffel words, so μ of the median is the product of the two matrices, and the Fricke identity $\text{tr}(XY) + \text{tr}(XY^{-1}) = \text{tr } X \text{tr } Y$ turns this into

$$m_{\text{median}} = 3m_{\alpha}m_{\beta} - m_{\text{other}}$$

for the three labels concerned. Every step of (52) therefore replaces the older vertex by the median of the two retained vertices. Hence the two Farey columns follow Fibonacci addition from $(1, 4)$ and $(1, 5)$. Their determinant has absolute value one, so all resulting fractions are reduced. Both coordinates grow without bound, and their ratios tend to (50). The subsequence at multiples of T has the same limit.

This limit lies strictly between $1/5$ and $1/4$. Its nearest critical slope is $1/5$, and

$$\rho - \frac{1}{5} = \frac{1}{5(4 + 5\Phi)} > 0.$$

For sufficiently large j , the distance of P_j/Q_j from Θ_{89} is at least half this positive number. Apply [corollary 7.3](#) with $\eta = 1/[10(4 + 5\Phi)]$ to obtain (51). $\square$

The argument needs neither the value of T nor a conjecture about prime values of recurrence sequences. It is explicit in the sense that the family is described by a rule and not merely shown to exist; the period T is large, so the members beyond the first are not small numbers. Its modular periodicity selects the required valuation, while the real Farey dynamics places the source slopes away from the critical set. These two elementary mechanisms make the quantitative theorem nonvacuous on genuine sources with unbounded numerators.

7.3. Why the equality slope alone is insufficient. The six-state equality graph has frequency $1/5$, but the full critical graph also contains the state

$$S = \begin{pmatrix} 101 & 19 \\ 19 & 82 \end{pmatrix} = \begin{pmatrix} 10 & 1 \\ 1 & 9 \end{pmatrix}^2.$$

This state commutes with $M(1)^2$. Exactly two forced row subtractions return it after one light paired letter, so the loop has zero cost. A fixed entry and primitive exit give the explicit family

$$q_j = \chi(21221122112 \ 1^j \ 2122121)2, \quad j \geq 0. \quad (53)$$

The prefix enters S with cumulative weight 15. The suffix reaches $(109, 49; 43, 92)$ with additional weight 20; its terminal vector is $89(3, 2)^T$ and its terminal charge is 1. Thus every member has exact norm valuation one at 89 and completed excess 36. Its inferred counts are $P = 21$ and $Q = 38 + 2j$. This rules out a cost bound proportional only to $|Q - 5P|$ on the full regular language.

Every displayed core contains both 11 and 22, so it is unbalanced. The family is consequently not a family of genuine Markoff sources and does not contradict [theorem 7.2](#). It explains why the zero-slack graph outside the exact equality language cannot simply be discarded when studying near-minimizers.

8. FURTHER QUESTIONS

The completed norm-89 comparison illustrates two distinct sources of rigidity. The endpoint potential makes cost nonnegative on a large regular language. Its equality graph then forces a count relation incompatible with a reduced genuine occurrence, except at the fixed base source. Finite balance adds a quantitative constraint: long paths with small completed excess must have frequency close to a recurrent zero-cost component.

Three questions remain particularly concrete. First, one can seek a description of endpoint potentials that persists as the Gaussian block varies. Any such statement must

retain both primitive content and the actual source and terminal charges. The present proof exhibits one explicit potential at 89 and supplies no potential uniform in the block.

Second, the graph frequencies $0, 1/5, 1/3, 1$ need not all be attainable as limits of genuine sources with bounded excess. The exact mechanical recognition condition is stronger than finite balance and coprime counts. Incorporating it into the critical-component analysis could remove some exceptional slopes or improve the constants. The current theorem makes neither assertion.

Third, minimization on an entire Gaussian fiber requires comparing allocations involving other prime blocks. Strict increase under one complete flip is a useful local statement about this arithmetic orbit, but it does not compare every primitive representation of the same norm. The infinite family in [theorem 7.4](#) provides genuine instances where this one-block penalty becomes large; understanding how several blocks interact remains a separate problem.

Two further directions were suggested to us, and both concern the finite graph rather than the arithmetic. The first is the decomposition used in the proof of [lemma 4.4](#), which cuts an accepted path at its positive-weight edges and treats each zero-weight segment separately. That is a first-return decomposition relative to the zero-slack subgraph, and the theory of return words gives a sharper vocabulary for such decompositions than the condensation budget ([31](#)) does: the budget charges every component of a longest condensation path, whereas a return-word induction would charge only the components a given word actually revisits. Whether that replaces the constant $K = 59$ by something close to the six-state component alone is the concrete question.

The second is the algebraic reading of finiteness recalled in [section 1](#). Over a finite field, a power series is algebraic exactly when a finite automaton generates its coefficients [[4](#)], and the reflection graph of [section 5](#) is a finite automaton attached to an arithmetic operation. The analogy stops there, since that equivalence is a positive-characteristic phenomenon and our labels are ordinary integers; we do not know whether some series formed from the costs Δ along the graph satisfies an algebraic relation, and we have not found a formulation in which the question is more than suggestive.

The general theorem in [theorem 4.5](#) can be used outside the Markoff setting. It takes as input a finite graph whose initial, edge and terminal weights are nonnegative integers and whose zero-weight cyclic components each carry a single letter frequency, and it returns an explicit bound on the letter count of every balanced word the graph accepts. Separating that mechanism from the arithmetic seems to us the most transferable part of this article.

SUPPLEMENTARY MATERIAL

The supplementary material accompanying this article contains the exact computations used in the proofs of [theorems 5.1, 6.1 and 7.2](#): the closed state graph, its table of potentials, the cyclic components of the zero-slack subgraph, and instructions for repeating the calculations. The reconstruction of the graph and of its potential was carried out twice, by implementations sharing no code. The material also carries independent confirmations of two results that are proved analytically above and take no computational input: the letter counts, the cost and the balance of the dictionary of [proposition 3.1](#), together with a direct evaluation of the excess from its definition, and the polynomial identities behind [theorem 7.4](#).

CODE AND DATA AVAILABILITY

The source code and the data supporting this article, with the instructions that repeat every calculation, are contained in the supplementary material of this article, which is available, together with that of the other articles of this series:

<https://github.com/dutykh/fibonacci-pell-nearest-gaps/>

REFERENCES

1. Jean Berstel and Michel Pocchiola, *A geometric proof of the enumeration formula for Sturmian words*, International Journal of Algebra and Computation **3** (1993), no. 3, 349–355.
2. Valérie Berthé, Aldo de Luca, and Christophe Reutenauer, *On an involution of Christoffel words and Sturmian morphisms*, European Journal of Combinatorics **29** (2008), no. 2, 535–553.
3. Jairo Bochi and Yiwei Zhang, *Ergodic optimization of prevalent super-continuous functions*, International Mathematics Research Notices **2016** (2016), no. 19, 5988–6017.
4. Gilles Christol, Teturo Kamae, Michel Mendès France, and Gérard Rauzy, *Suites algébriques, automates et substitutions*, Bulletin de la Société Mathématique de France **108** (1980), 401–419.
5. Brandon Dong, Soren Dupont, Evan M. O’Dorney, and W. Theo Waitkus, *Raney transducers and the lowest point of the n -Lagrange spectrum*, INTEGERS **25** (2025), A47, 30 pp.
6. Jacques Justin and Laurent Vuillon, *Return words in Sturmian and episturmian words*, RAIRO – Theoretical Informatics and Applications **34** (2000), no. 5, 343–356.
7. M. Lothaire, *Algebraic combinatorics on words*, Encyclopedia of Mathematics and its Applications, vol. 90, Cambridge University Press, Cambridge, 2002.
8. Filippo Mignosi, *On the number of factors of Sturmian words*, Theoretical Computer Science **82** (1991), no. 1, 71–84.
9. Mehryar Mohri, *Weighted automata algorithms*, Handbook of Weighted Automata (Manfred Droste, Werner Kuich, and Heiko Vogler, eds.), Monographs in Theoretical Computer Science. An EATCS Series, Springer, Berlin, Heidelberg, 2009, pp. 213–254.
10. J. P. Quadrat and Max-Plus Working Group, *Min-plus linearity and statistical mechanics*, Markov Processes and Related Fields **3** (1997), no. 4, 565–587.
11. George N. Raney, *On continued fractions and finite automata*, Mathematische Annalen **206** (1973), 265–283.
12. Christophe Reutenauer, *Christoffel words and Markoff triples*, INTEGERS **9** (2009), no. 3, 327–332.
13. Laurent Vuillon, *A characterization of Sturmian words by return words*, European Journal of Combinatorics **22** (2001), no. 2, 263–275.

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